

Polynitrogen Allotropes

Screened structures: energies and depictions

Neutral, anionic and cationic N_x clusters

ground states and metastable isomers

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Generated by the polyN pipeline

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How to read this report

Structures are grouped by charge family, then by cluster size N_x . Within each series, isomers are ordered by increasing energy; the most stable one (the *ground state*, GS) is highlighted and defines the zero of the **relative energy** ΔE_{rel} (kcal/mol). The **reaction energy** E_{react} (kcal/mol) is computed against the pipeline's per-family reference reaction:

- neutral: $N_x \rightarrow x/2 N_2$;
- cation: $N_x^+ \rightarrow x/2 N_2 + N_5^+$;
- anion: $N_x^- \rightarrow x/2 N_2 + N_5^-$.

Each `.xyz` filename refers to a clean, VESTA-readable file in the `xyz_clean/` folder alongside this report. All energies are at the GFN2-xTB level; structures are frequency-verified true minima and passed the fragmentation filter.

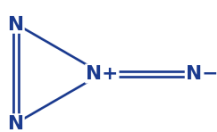
1. Neutral family

N₄

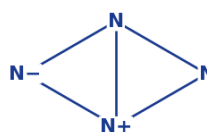
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1 (GS)	N4_neutral_neutral_001.xyz	0.0	+27.3
2	N4_neutral_neutral_002.xyz	26.5	+53.8
3	N4_neutral_neutral_003.xyz	31.0	+58.3
4	N4_neutral_neutral_004.xyz	41.6	+69.0



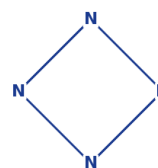
#1



#2



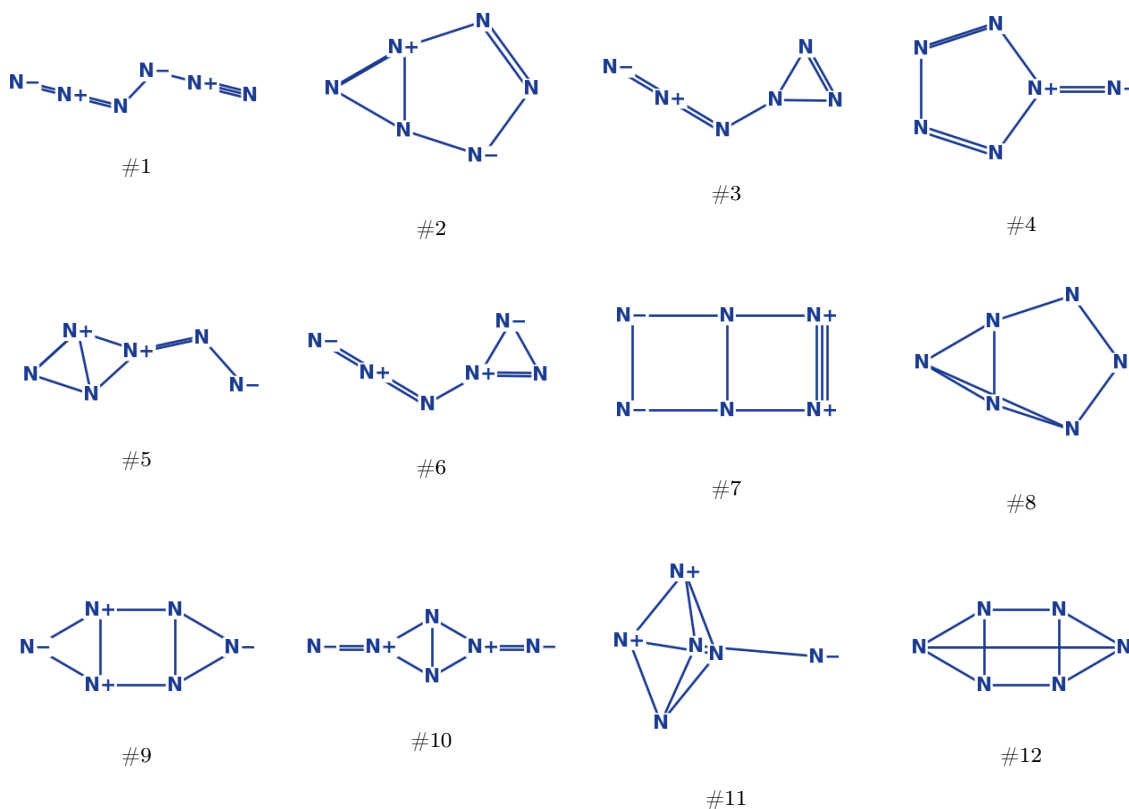
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#4

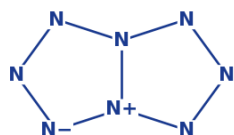
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3	N6_neutral_neutral_003.xyz	55.0	-8.7
4	N6_neutral_neutral_004.xyz	60.6	-3.1
5	N6_neutral_neutral_005.xyz	64.6	+0.9
6	N6_neutral_neutral_006.xyz	78.8	+15.1
7	N6_neutral_neutral_007.xyz	83.6	+19.9
8	N6_neutral_neutral_008.xyz	84.5	+20.8
9	N6_neutral_neutral_009.xyz	110.8	+47.1
10	N6_neutral_neutral_010.xyz	114.3	+50.6
11	N6_neutral_neutral_011.xyz	119.6	+55.9
12	N6_neutral_neutral_012.xyz	136.7	+73.0

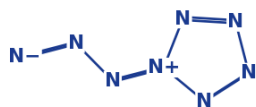


N_8

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3	N8_neutral_neutral_003.xyz	20.3	-93.1
4	N8_neutral_neutral_004.xyz	28.1	-85.3
5	N8_neutral_neutral_005.xyz	29.4	-84.0
6	N8_neutral_neutral_006.xyz	56.5	-56.9
7	N8_neutral_neutral_007.xyz	66.4	-47.0
8	N8_neutral_neutral_008.xyz	67.2	-46.2
9	N8_neutral_neutral_009.xyz	73.7	-39.7
10	N8_neutral_neutral_010.xyz	84.1	-29.3
11	N8_neutral_neutral_011.xyz	84.8	-28.6
12	N8_neutral_neutral_012.xyz	94.0	-19.4
13	N8_neutral_neutral_013.xyz	102.9	-10.5
14	N8_neutral_neutral_014.xyz	113.2	-0.1
15	N8_neutral_neutral_015.xyz	126.3	+12.9
16	N8_neutral_neutral_016.xyz	132.8	+19.4
17	N8_neutral_neutral_017.xyz	139.0	+25.7
18	N8_neutral_neutral_018.xyz	140.2	+26.8
19	N8_neutral_neutral_019.xyz	150.3	+36.9
20	N8_neutral_neutral_020.xyz	152.9	+39.5
21	N8_neutral_neutral_021.xyz	155.7	+42.3
22	N8_neutral_neutral_022.xyz	161.9	+48.5
23	N8_neutral_neutral_023.xyz	164.7	+51.3
24	N8_neutral_neutral_024.xyz	183.1	+69.7
25	N8_neutral_neutral_025.xyz	205.7	+92.3



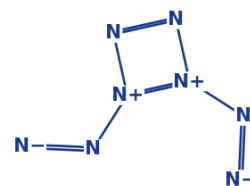
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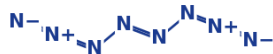
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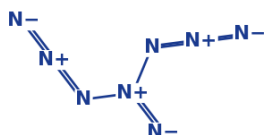
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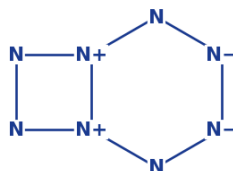
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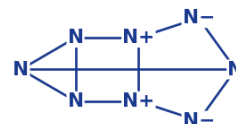
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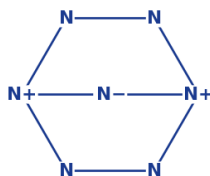
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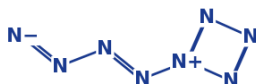
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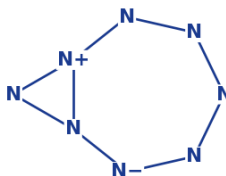
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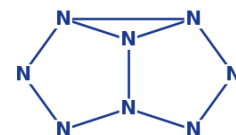
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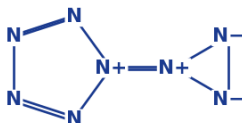
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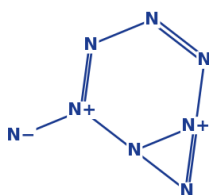
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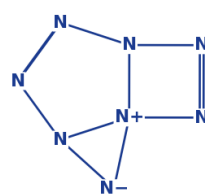
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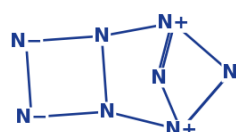
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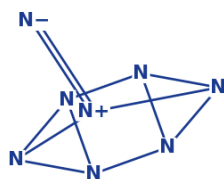
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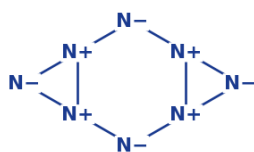
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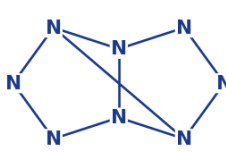
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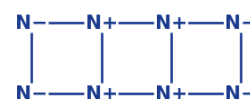
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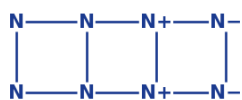
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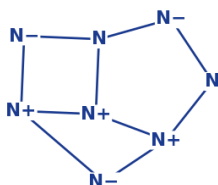
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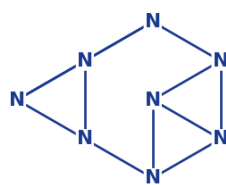
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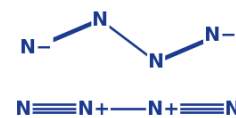
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#22



#23



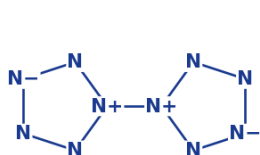
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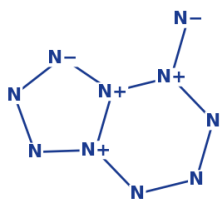
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N_{10}

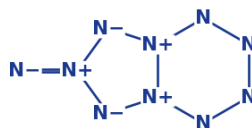
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2	N10_neutral_neutral_002.xyz	10.5	-123.7
3	N10_neutral_neutral_003.xyz	23.5	-110.7
4	N10_neutral_neutral_004.xyz	33.4	-100.8
5	N10_neutral_neutral_005.xyz	36.6	-97.6
6	N10_neutral_neutral_006.xyz	38.6	-95.6
7	N10_neutral_neutral_007.xyz	53.7	-80.6
8	N10_neutral_neutral_008.xyz	59.3	-74.9
9	N10_neutral_neutral_009.xyz	63.5	-70.7
10	N10_neutral_neutral_010.xyz	88.6	-45.6
11	N10_neutral_neutral_011.xyz	93.0	-41.2
12	N10_neutral_neutral_012.xyz	98.8	-35.4
13	N10_neutral_neutral_013.xyz	99.6	-34.6
14	N10_neutral_neutral_014.xyz	100.5	-33.7
15	N10_neutral_neutral_015.xyz	111.2	-23.0
16	N10_neutral_neutral_016.xyz	113.8	-20.4
17	N10_neutral_neutral_017.xyz	130.5	-3.7
18	N10_neutral_neutral_018.xyz	143.0	+8.8
19	N10_neutral_neutral_019.xyz	148.0	+13.8
20	N10_neutral_neutral_020.xyz	168.5	+34.3
21	N10_neutral_neutral_021.xyz	174.6	+40.4
22	N10_neutral_neutral_022.xyz	183.9	+49.7



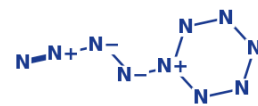
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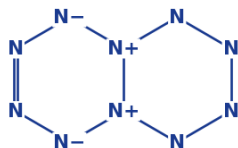
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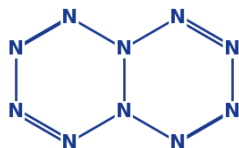
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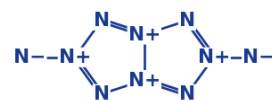
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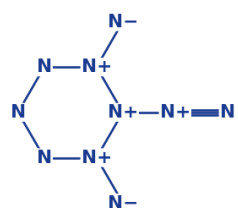
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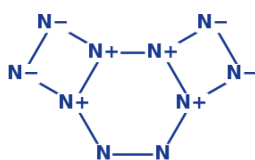
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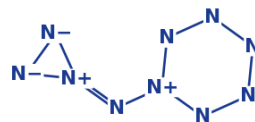
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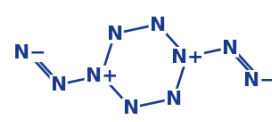
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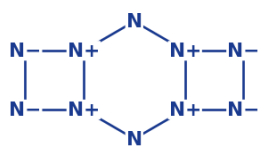
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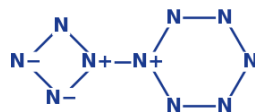
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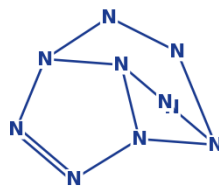
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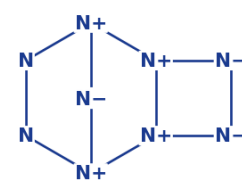
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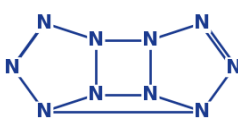
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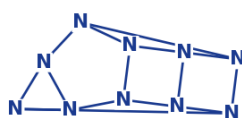
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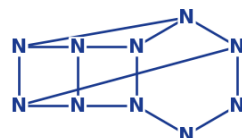
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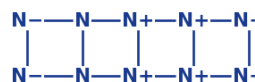
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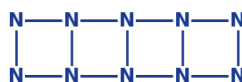
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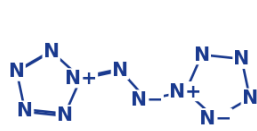
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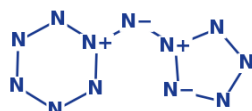
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N_{12}

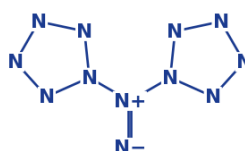
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2	N12_neutral_neutral_002.xyz	13.2	-134.7
3	N12_neutral_neutral_003.xyz	35.9	-112.0
4	N12_neutral_neutral_004.xyz	45.6	-102.3
5	N12_neutral_neutral_005.xyz	68.5	-79.4
6	N12_neutral_neutral_006.xyz	76.4	-71.6
7	N12_neutral_neutral_007.xyz	101.0	-47.0
8	N12_neutral_neutral_008.xyz	133.2	-14.8
9	N12_neutral_neutral_009.xyz	242.4	+94.5



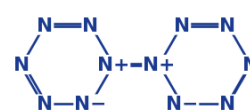
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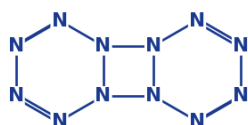
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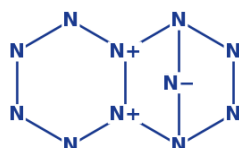
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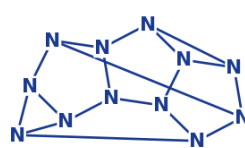
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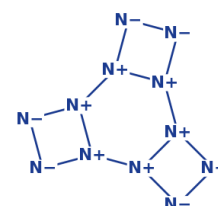
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#6



#7



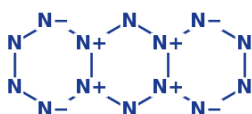
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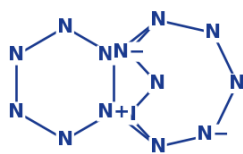
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N_{14}

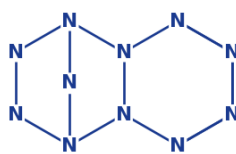
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1 (GS)	N14_neutral_neutral_001.xyz	0.0	-140.2
2	N14_neutral_neutral_002.xyz	40.8	-99.5
3	N14_neutral_neutral_003.xyz	77.9	-62.3
4	N14_neutral_neutral_004.xyz	101.6	-38.6
5	N14_neutral_neutral_005.xyz	133.3	-6.9



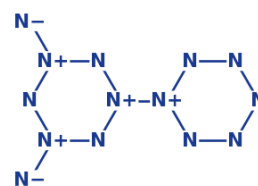
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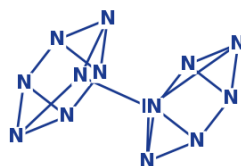
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#3



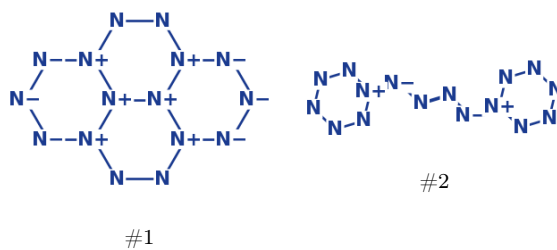
#4



#5

N_{16}

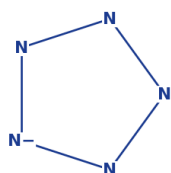
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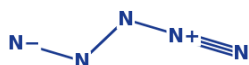
2. Anionic family

 N_5^-

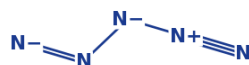
Rank	.xyz file	ΔE_{rel} (kcal/mol)	E_{react} (kcal/mol)
1 (GS)	N5_anion_anion_001.xyz	0.0	-0.0
2	N5_anion_anion_002.xyz	45.0	+45.0
3	N5_anion_anion_003.xyz	45.8	+45.8
4	N5_anion_anion_004.xyz	85.9	+85.9
5	N5_anion_anion_005.xyz	101.1	+101.1
6	N5_anion_anion_006.xyz	104.3	+104.3
7	N5_anion_anion_007.xyz	110.2	+110.2
8	N5_anion_anion_008.xyz	118.6	+118.6
9	N5_anion_anion_009.xyz	119.7	+119.7
10	N5_anion_anion_010.xyz	143.4	+143.4



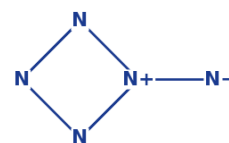
#1



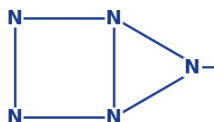
#2



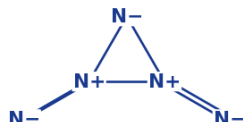
#3



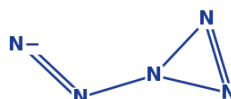
#4



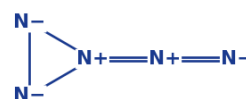
#5



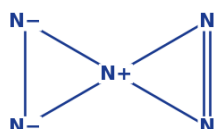
#6



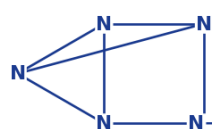
#7



#8



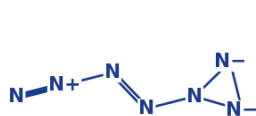
#9



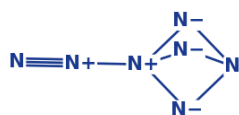
#10

N_7^-

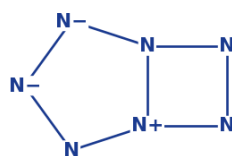
Rank	.xyz file	ΔE_{rel} (kcal/mol)	E_{react} (kcal/mol)
1 (GS)	N7_anion_anion_001.xyz	0.0	+2.1
2	N7_anion_anion_002.xyz	2.8	+4.9
3	N7_anion_anion_003.xyz	7.9	+10.0
4	N7_anion_anion_004.xyz	9.1	+11.2
5	N7_anion_anion_005.xyz	22.2	+24.3
6	N7_anion_anion_006.xyz	30.9	+32.9
7	N7_anion_anion_007.xyz	41.7	+43.7
8	N7_anion_anion_008.xyz	46.2	+48.3
9	N7_anion_anion_009.xyz	47.5	+49.5
10	N7_anion_anion_010.xyz	50.4	+52.5
11	N7_anion_anion_011.xyz	51.7	+53.8
12	N7_anion_anion_012.xyz	52.3	+54.4
13	N7_anion_anion_013.xyz	54.1	+56.2
14	N7_anion_anion_014.xyz	59.9	+62.0
15	N7_anion_anion_015.xyz	71.6	+73.7
16	N7_anion_anion_016.xyz	74.3	+76.4
17	N7_anion_anion_017.xyz	82.0	+84.1
18	N7_anion_anion_018.xyz	97.0	+99.0
19	N7_anion_anion_019.xyz	102.4	+104.4
20	N7_anion_anion_020.xyz	144.9	+147.0



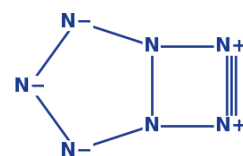
#1



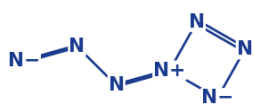
#2



#3



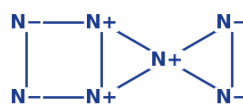
#4



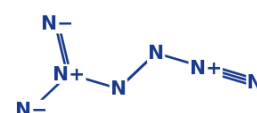
#5



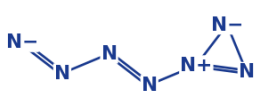
#6



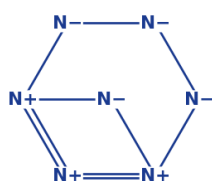
#7



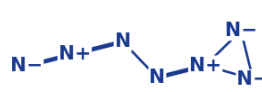
#8



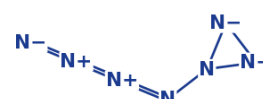
#9



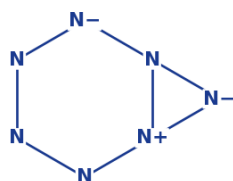
#10



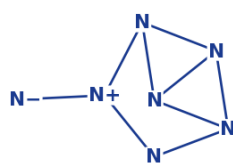
#11



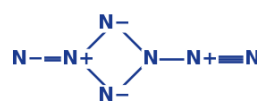
#12



#13



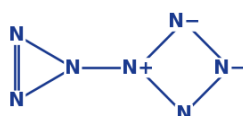
#14



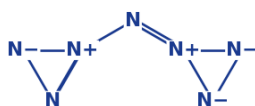
#15



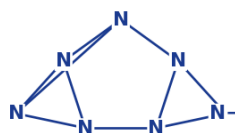
#16



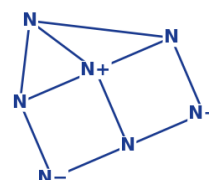
#17



#18



#19



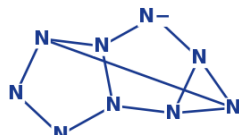
#20

N_9^-

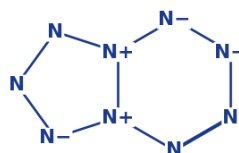
Rank	.xyz file	ΔE_{rel} (kcal/mol)	E_{react} (kcal/mol)
1 (GS)	N9_anion_anion_001.xyz	0.0	-67.2
2	N9_anion_anion_002.xyz	19.4	-47.9
3	N9_anion_anion_003.xyz	23.6	-43.6
4	N9_anion_anion_004.xyz	29.5	-37.8
5	N9_anion_anion_005.xyz	33.1	-34.2
6	N9_anion_anion_006.xyz	35.9	-31.3
7	N9_anion_anion_007.xyz	46.4	-20.9
8	N9_anion_anion_008.xyz	53.9	-13.3
9	N9_anion_anion_009.xyz	55.9	-11.3
10	N9_anion_anion_010.xyz	57.1	-10.2
11	N9_anion_anion_011.xyz	60.0	-7.2
12	N9_anion_anion_012.xyz	86.8	+19.6
13	N9_anion_anion_013.xyz	87.4	+20.2
14	N9_anion_anion_014.xyz	88.4	+21.2
15	N9_anion_anion_015.xyz	105.0	+37.7
16	N9_anion_anion_016.xyz	126.1	+58.8
17	N9_anion_anion_017.xyz	128.0	+60.8
18	N9_anion_anion_018.xyz	141.3	+74.1
19	N9_anion_anion_019.xyz	154.0	+86.7
20	N9_anion_anion_020.xyz	205.4	+138.2



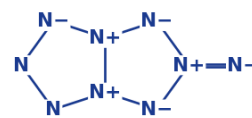
#1



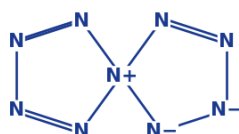
#2



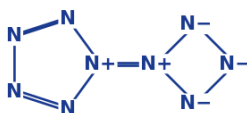
#3



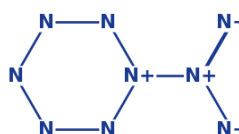
#4



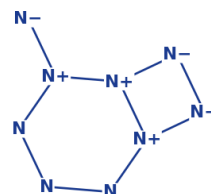
#5



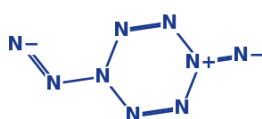
#6



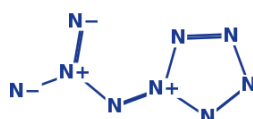
#7



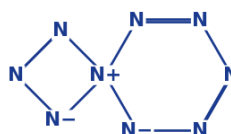
#8



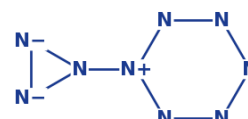
#9



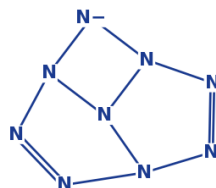
#10



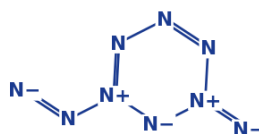
#11



#12



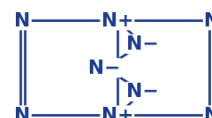
#13



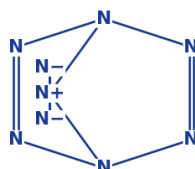
#14



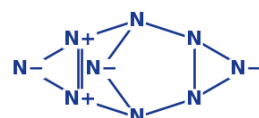
#15



#16



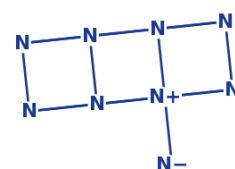
#17



#18



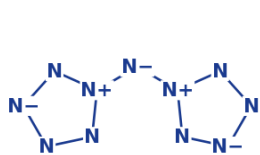
#19



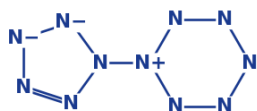
#20

N_{11}^-

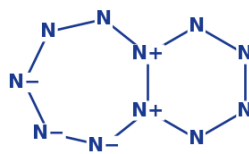
Rank	.xyz file	ΔE_{rel} (kcal/mol)	E_{react} (kcal/mol)
1 (GS)	N11_anion_anion_001.xyz	0.0	-126.2
2	N11_anion_anion_002.xyz	39.4	-86.7
3	N11_anion_anion_003.xyz	44.2	-82.0
4	N11_anion_anion_004.xyz	49.8	-76.4
5	N11_anion_anion_005.xyz	51.6	-74.5
6	N11_anion_anion_006.xyz	57.4	-68.8
7	N11_anion_anion_007.xyz	69.8	-56.4
8	N11_anion_anion_008.xyz	77.4	-48.8
9	N11_anion_anion_009.xyz	79.3	-46.9
10	N11_anion_anion_010.xyz	89.5	-36.7
11	N11_anion_anion_011.xyz	92.7	-33.5
12	N11_anion_anion_012.xyz	108.7	-17.5
13	N11_anion_anion_013.xyz	115.9	-10.3
14	N11_anion_anion_014.xyz	119.7	-6.5
15	N11_anion_anion_015.xyz	146.0	+19.8
16	N11_anion_anion_016.xyz	164.0	+37.8
17	N11_anion_anion_017.xyz	164.9	+38.7
18	N11_anion_anion_018.xyz	185.2	+59.0



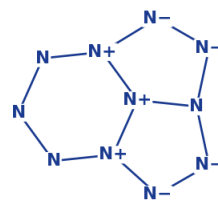
#1



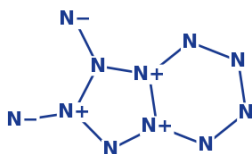
#2



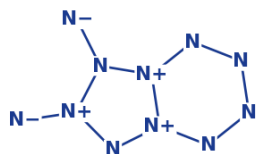
#3



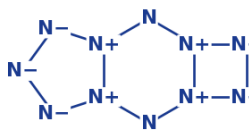
#4



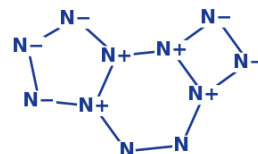
#5



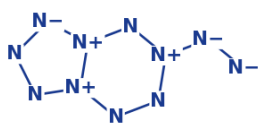
#6



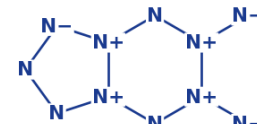
#7



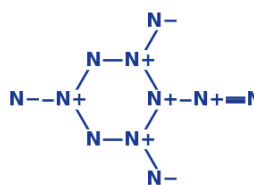
#8



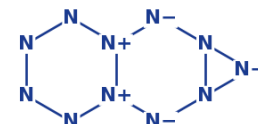
#9



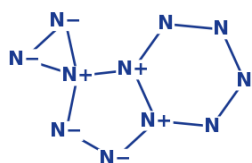
#10



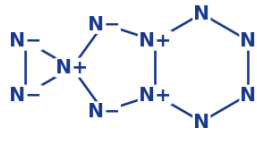
#11



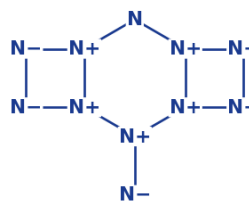
#12



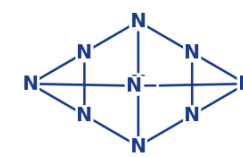
#13



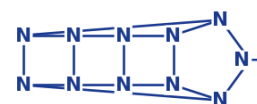
#14



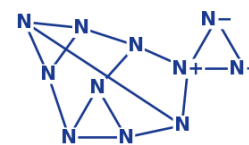
#15



#16



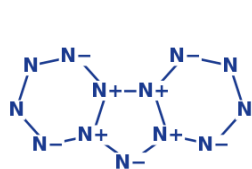
#17



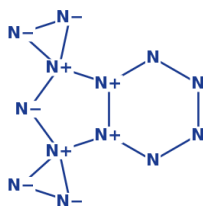
#18

N_{13}^-

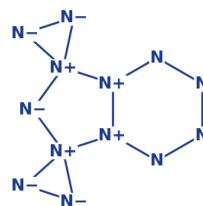
Rank	.xyz file	ΔE_{rel} (kcal/mol)	E_{react} (kcal/mol)
1 (GS)	N13_anion_anion_001.xyz	0.0	-92.6
2	N13_anion_anion_002.xyz	115.0	+22.4
3	N13_anion_anion_003.xyz	116.6	+24.0



#1



#2

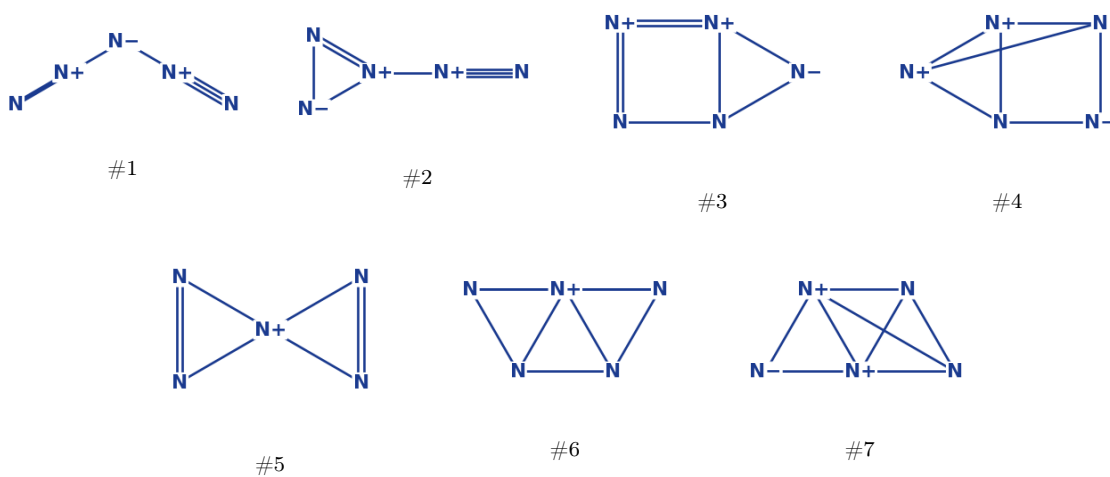


#3

3. Cationic family

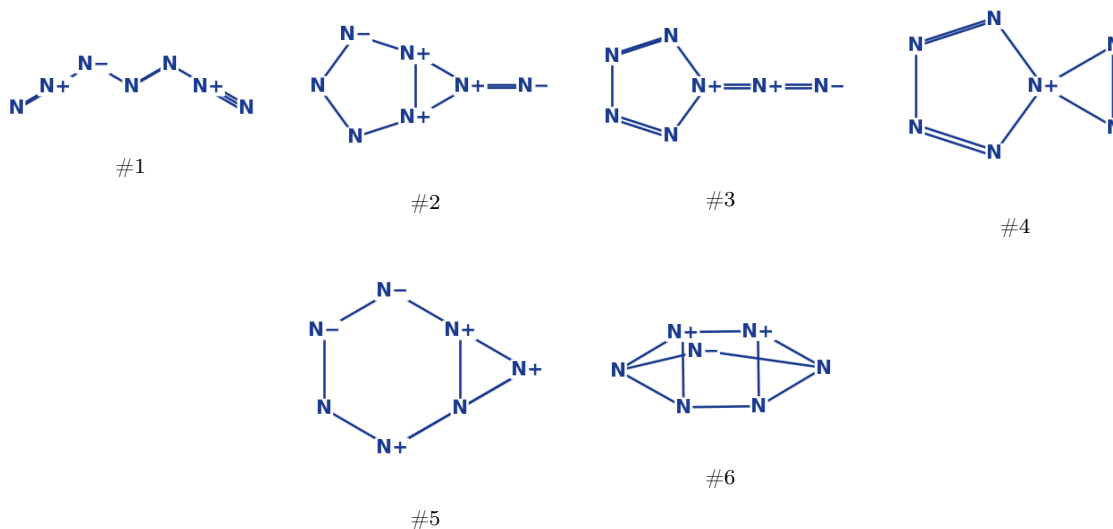
 N_5^+

Rank	.xyz file	ΔE_{rel} (kcal/mol)	E_{react} (kcal/mol)
1 (GS)	N5_cation_cation_001.xyz	0.0	-105.6
2	N5_cation_cation_002.xyz	74.2	-31.5
3	N5_cation_cation_003.xyz	104.8	-0.8
4	N5_cation_cation_004.xyz	105.6	+0.0
5	N5_cation_cation_005.xyz	134.0	+28.4
6	N5_cation_cation_006.xyz	184.3	+78.7
7	N5_cation_cation_007.xyz	185.6	+80.0



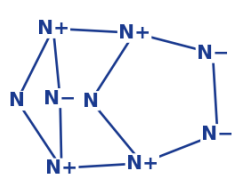
N_7^+

Rank	.xyz file	ΔE_{rel} (kcal/mol)	E_{react} (kcal/mol)
1 (GS)	N7_cation_cation_001.xyz	0.0	-133.4
2	N7_cation_cation_002.xyz	6.2	-127.2
3	N7_cation_cation_003.xyz	27.0	-106.4
4	N7_cation_cation_004.xyz	91.3	-42.2
5	N7_cation_cation_005.xyz	124.9	-8.5
6	N7_cation_cation_006.xyz	178.8	+45.4

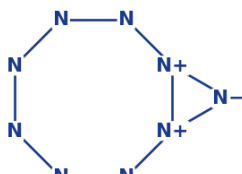


N_9^+

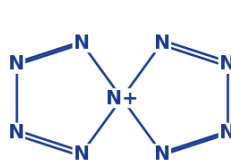
Rank	.xyz file	ΔE_{rel} (kcal/mol)	E_{react} (kcal/mol)
1 (GS)	N9_cation_cation_001.xyz	0.0	-174.4
2	N9_cation_cation_002.xyz	29.4	-145.0
3	N9_cation_cation_003.xyz	32.0	-142.5
4	N9_cation_cation_004.xyz	34.2	-140.2
5	N9_cation_cation_005.xyz	100.6	-73.8
6	N9_cation_cation_006.xyz	104.5	-69.9
7	N9_cation_cation_007.xyz	113.9	-60.6
8	N9_cation_cation_008.xyz	114.9	-59.5
9	N9_cation_cation_009.xyz	117.3	-57.1
10	N9_cation_cation_010.xyz	132.3	-42.1
11	N9_cation_cation_011.xyz	174.2	-0.3
12	N9_cation_cation_012.xyz	176.8	+2.4



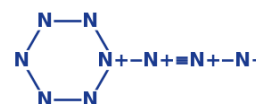
#1



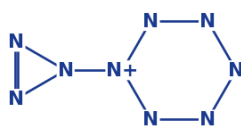
#2



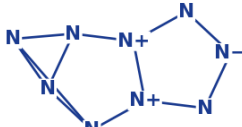
#3



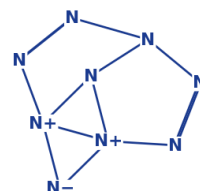
#4



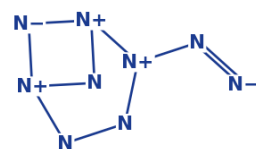
#5



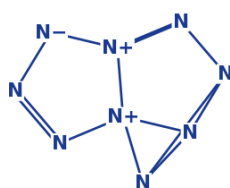
#6



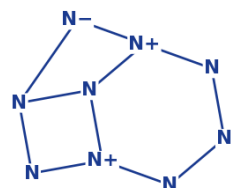
#7



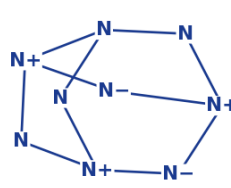
#8



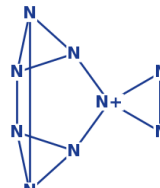
#9



#10



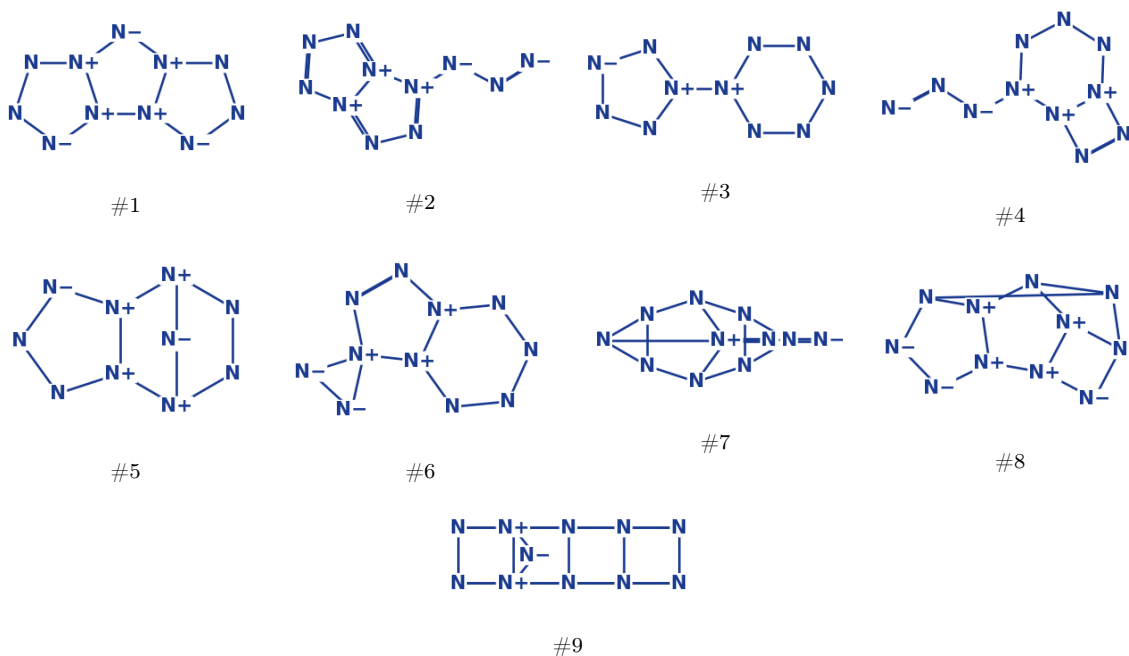
#11



#12

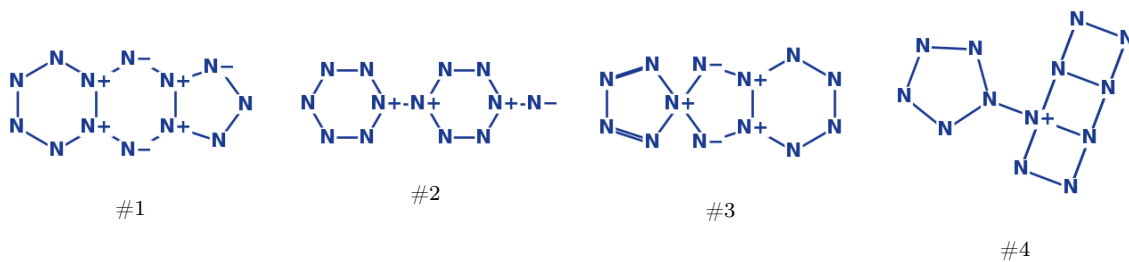
N_{11}^+

Rank	.xyz file	ΔE_{rel} (kcal/mol)	E_{react} (kcal/mol)
1 (GS)	N11_cation_cation_001.xyz	0.0	-200.5
2	N11_cation_cation_002.xyz	2.7	-197.9
3	N11_cation_cation_003.xyz	46.1	-154.4
4	N11_cation_cation_004.xyz	61.4	-139.2
5	N11_cation_cation_005.xyz	73.6	-126.9
6	N11_cation_cation_006.xyz	100.3	-100.2
7	N11_cation_cation_007.xyz	128.1	-72.4
8	N11_cation_cation_008.xyz	195.4	-5.1
9	N11_cation_cation_009.xyz	239.7	+39.2



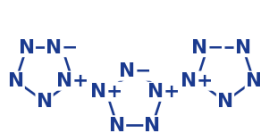
N_{13}^+

Rank	.xyz file	ΔE_{rel} (kcal/mol)	E_{react} (kcal/mol)
1 (GS)	N13_cation_cation_001.xyz	0.0	-213.3
2	N13_cation_cation_002.xyz	40.5	-172.9
3	N13_cation_cation_003.xyz	56.5	-156.8
4	N13_cation_cation_004.xyz	194.1	-19.2

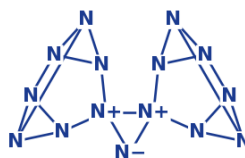


N_{15}^+

Rank	.xyz file	ΔE_{rel} (kcal/mol)	E_{react} (kcal/mol)
1 (GS)	N15_cation_cation_001.xyz	0.0	-252.5
2	N15_cation_cation_002.xyz	259.3	+6.8



#1



#2